

Harshit (PhD | M.Tech | B.Arch)

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Education

- [2018 - 2024] **Doctor of Philosophy from Indian Institute of Technology Roorkee, India**
 - ◆ Digital Twin Modelling, 3D GIS, Photogrammetric Computer Vision, Point Cloud Processing, 3D GIS Applications, Open-Source Development, UAV-Based 3D Reconstruction, 3D Geospatial Modelling [Python, CGAL, Pix4DMapper, Metashape, 3D Survey, CloudCompare, Meshlab]
- [2015 - 2027] **M.Tech from Indian Institute of Technology Roorkee, India**
 - ◆ UAV Photogrammetry, 3D reconstruction, Geomatics, GIS & Remote Sensing, Spatial Analysis, Geospatial Modelling, DEM processing, Terrain analysis, Digital Image Processing, GPS, Geodesy. [Metashape, Pix4D, ArcGIS, ERDAS, ENVI, QGIS]
- [2009 - 2014] **B.Arch from National Institute of Technology Patna, India**
 - ◆ Museum + Exhibition Planning, Architectural Design, BIM & CAD, Construction Knowledge, Architectural Design Principles, Building Services, Urban Design, Structural and Working Drawings, 3D Modelling and Visualization, Architectural Graphics, Disaster Resistant Architecture, Architectural Thesis Presentation. [Autodesk (CAD & Revit), Trimble SketchUp, Adobe Photoshop]

Experience

- [2024 - 2026] **Postdoc Researcher (Data Science Section, IT University of Copenhagen)**
 - ◆ Neural Rendering (NeRF & Gaussian Splatting), Dynamic Scene Modeling (4D / temporal coherence), Computer Vision & View Synthesis, Mixed Reality & Immersive Systems, Portable Multi-view Gaussian frame using GoPro Hero 13 Cameras
- [2020 - 2024] **Senior Research Fellow & Teaching Assistant (Geospatial Engineering Group, IIT Roorkee)**
 - ◆ Photogrammetry & 3D Reconstruction, Teaching Assistance, Geospatial Engineering Education, Surveying Supervision, Field Training, Academic Support
 - ◆ Hands-on Experience with DJI Matrice 300 series with Zenmuse L1, Phantom 4 Pro, DJI Inspire 2, Faro Lidar, Apple LiDAR, Trimble Total station, MicaSense MSS, Spectra and Sokkia GPS.
- [2017 - 2018] **Research Associate (Sponsored Research (SR) and Industrial Consultancy (IC), IIT Roorkee, India)**
 - ◆ Consultancy services, Monitoring Fly Ash ponds inundation, Eccentricity Assessment of as-built Infrastructures.
- [2014 - 2015] **Architect (Free-Lancing)**
 - ◆ Architectural Design, Mixed-Use Development, Sustainable Design, Project Coordination, Construction Documentation

Grants & Awards

- The ISPRS Foundation (TIF) Award [Geospatial Week 2023]
- Travel Grant / Award [IEEE IGARSS 2024, 3D Geoinfo 2022],
- Travel support [Office of Dean of Resources and Alumni Affairs (DORA) IIT Roorkee, 2022]
- Department Development Fund [Department of Civil Engineering, IIT Roorkee, 2022]
- IEEE IFT UAV Competition "Instrumentation and Data Processing Trophy 2021"

Languages: Hindi (Mother tongue), English (Professional proficiency), Bengali, Sanskrit

Creative Skills: *Music:* playing flute & guitar, writing & singing songs. *Literature:* Reading philosophical investigations and scientific revelations. *Sports:* Badminton, Cycling, Running, Cricket and Squash. *Trekking:* summit done for Triund Trek (Altitude 2875m), Devariya-taal (Altitude 2438m), Chandrashila Trek (Altitude 4090m)